

Phase Variable Module for the 2nd Generation Open-Source Leg

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This code release contains the python script and a compiled library for running the Locolab's Phase Variable Module on the second generation [Open-Source Leg \(OSL\)](#). This module is a thigh-based phase variable that continuously estimates the user's progression throughout the gait cycle. The phase variable is calculated using a piecewise-linear mapping of the user's global thigh angle as detected by an Inertial Measurement Unit. This example script shows how to use the module on the OSL, including reading from an inertial thigh sensor, calling the phase estimation algorithm, and logging data.

The compiled controller implements the control logic detailed in the following publications:

- T. K. Best, C. G. Welker, E. J. Rouse and R. D. Gregg, "Data-Driven Variable Impedance Control of a Powered Knee-Ankle Prosthesis for Adaptive Speed and Incline Walking," in *IEEE Transactions on Robotics*, vol. 39, no. 3, pp. 2151-2169, June 2023, doi: 10.1109/TRO.2022.3226887.

We kindly request that you cite this publication in any research projects that use this code.

License

This library is released under the [Polyform Strict License](#).

Dependencies

The following packages are required to run this script:

- numpy
- opensourceleg (available via `pip install`)
- MSCL: The Microstrain Community Library. You can find installation instructions on the [MSCL GitHub Page](#). For a quick installation, run the following commands:

- `wget https://github.com/LORD-MicroStrain/MSCL/releases/latest/download/python3-mscl_65.0.0_armhf.deb`
- `sudo apt install ./python3-mscl_65.0.0_armhf.deb`
- If you would rather, you may substitute the Microstrain IMU for some other means of calculating the global thigh angle and velocity and no longer require this dependency. See below for details.

Using The Library

Note: This module does not include a walking controller. Rather, this phase variable algorithm should be paired with an existing walking controller, such as the examples on the Open-Source Leg website.

Prerequisites:

1. Ensure that all dependencies are installed.
2. Wire the IMU to the Pi via either UART or a USB to UART converter.
3. If applicable, enable the UART port on the Pi in `/boot/config.txt`, set `enable_uart=1`. Alternatively, you can enable the UART in `sudo raspi-config`.
4. Place the main python file and the compiled library in the same directory¹ on the OSL's raspberry pi.
5. Connect to the raspberry pi via ssh. We typically use Visual Studio Code as recommended by the Open-Source Leg Community.

Basic Usage:

1. Start by running the phase variable python script. The system will then zero the loadcell. Make sure the foot is off the ground and the shank is pointing straight up and down when zeroing the loadcell.
2. The phase variable module repeatedly writes to the terminal with:
 - a. The estimated gait phase (ranging from 0 to 1)
 - b. The estimated stance phase (ranging from 0 to 1)
 - c. The estimated swing phase (ranging from 0 to 1)
 - d. The state of the phase variable state machine (ranging from 0 to 6; states 1-3 occur during stance and states 4-6 occur during swing)

¹ If you would rather keep the compiled library in a different location, you must update the library path in the `CompiledController` constructor.

3. To end the main loop, press ctrl+c in the terminal.

Configuration

There is a configuration block at the top of the python script that allows you to change the phase estimation settings. You can modify the calibration matrix of your OSL's loadcell matrix, loop frequency, logger name, and the user's task.

Loadcell Calibration Matrix

You must replace the loadcell calibration matrix with the calibration matrix for your specific hardware. See your loadcell datasheet for details.

Loop Frequency

The phase variable is set with a loop frequency of 300 Hz but can be changed if desired.

Logger Name

You may customize the logger name (the default logger name is 'phase_var').

Setting the Task

In this example code, you may change the user's task: ramp incline (from -10 degrees to +10 degrees) and walking speed (ranging from 0.8 to 1.2 m/s). These values do not need to be constants, and can be changed in real-time.

Using a Different Method for Global Thigh Angle Measurement

You may use another means to measure the global thigh angle if you would rather. For example, some research groups already have an Xsens system, and therefore prefer to use it instead of the Microstrain IMU. Note that the controller's behavior depends greatly on this signal, and it is very important to provide high-quality measurements of the thigh angle and velocity. Noisy or drifting values may cause undesired behaviors.

To use a different sensor, simply replace the writes to `PhaseVariable.inputs.thighAngle_rad` and `PhaseVariable.inputs.thighVelocity_rad_s` with the appropriate readings from your sensor within the main loop.

Required Inputs

This script takes inputs from the inertial thigh sensor of the thigh angle in degrees and the thigh velocity in degrees per second. The vertical ground reaction force is directly measured by the OSL's six-axis load cell. Walking incline and speed are inputs required by the user before executing the phase variable code.

Contact

Questions regarding the control release should be posted on the Open-Source Leg Forum. Posting on the forum allows the community to benefit from our discussions. For other inquiries regarding our research, please contact Dr. Robert Gregg at rdgregg@umich.edu.

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